

A Data-Driven Approach to Support Local Business Recovery in Shopping Malls

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Abstract—Following the COVID-19 pandemic, consumer behavior has substantially changed in physical shopping centers, creating a challenging commercial landscape where small enterprises heavily struggle to adjust to decreased spending habits and heightened competition from big-box retail chains. This document suggests a data-driven system architecture designed to assist local business recovery through an analytical platform that optimizes client interaction, captures environmental variables, and automates key elements of inventory control. The outcomes of the empirical observation and structured surveys show that the primary operational barrier is not an absolute absence of clients, but rather a lack of consolidated and accessible data that small businesses can leverage to enhance real-time decision-making and expand sales conversion prospects.

I. INTRODUCTION

In recent years, shopping malls have evolved from being purely commercial spaces into multifunctional environments where people not only shop, but also socialize, eat, work, and spend their leisure time. This transformation became even more evident after the COVID-19 pandemic, which significantly changed consumer behavior and expectations. Studies such as [1], [2] show that customers now prioritize experience, convenience, and value over traditional purchasing habits. Additionally, many consumers combine online and physical shopping, creating hybrid consumption patterns that challenge traditional retail models [3], [4]. This shift forces businesses to rethink how they interact with customers and how they manage their operations.

Large retail chains have adapted effectively to these changes by adopting advanced data-driven strategies. Among them is the widespread application of digital platforms, integrated inventory systems, and predictive analytics that enable them to comprehend client preferences, forecast demand, and streamline complex processes [5]. These capacities provide them with a sizable competitive edge and enable them to react swiftly to volatile shifts in the market environment.

In contrast, small businesses face critical limitations that threaten their long-term economic sustainability. Many of them still rely on manual processes or fragmented, outdated sources of information, which reduces their capability to respond to customer needs in real time. As a result, they experience persistent operational problems such as low sales conversion, inefficient inventory management, and a gradual loss of loyal

customers to larger competitors [6]. Another important factor is the structural change in spending behavior, given that a considerable percentage of customers reported reducing their overall non-essential expenses after the global health crisis. This direct financial impact highlights the importance of making accurate, informed, and timely data choices within real-world commercial centers.

This creates an analytical gap between changing customer demand and physical business supply. While customers are physically present in shopping malls, their purchasing needs are not always satisfied due to poor information management and an inability to track trends dynamically. Understanding this gap is essential to improving the performance, resilience, and sustainability of small businesses. In this paper, we analyze the behavior of customers and small businesses within a real shopping mall environment. Based on this comprehensive analysis, we propose a modular data-driven system that integrates customer behavior, environmental variables, and business data to actively support tactical decision-making.

II. METHODS AND MATERIALS

To understand the core problem, a strategic combination of qualitative and quantitative methods was used, ensuring enough technical detail to allow complete replication of the research approach. Direct observation was deployed to analyze how customers move within the shopping mall layout. This field method allowed identifying high-traffic areas, peak hours, and specific spatial patterns of behavior, helping detect zones where high customer presence does not translate into effective sales conversion.

Complementing this field overview, customers and store employees were surveyed to extract distinct perspectives. Consumers gave precise details about their purchasing habits, frequency of visits, and individual choices, while workers shared sensitive information about inventory management problems, daily sales patterns, and operational difficulties. The baseline analysis was enhanced by this blend of viewpoints. Furthermore, the environmental analysis focused on external variables such as weather conditions, day of the week, and special calendar events. These factors influence customer behavior and introduce heavy variability into the retail environment, making business decision-making more complex.

A. System Representation

The overall conceptual mechanics of the solution are illustrated in the general representation of the system shown in Fig. 1. This diagram captures how external environmental inputs filter through the commercial infrastructure to influence consumer choices.

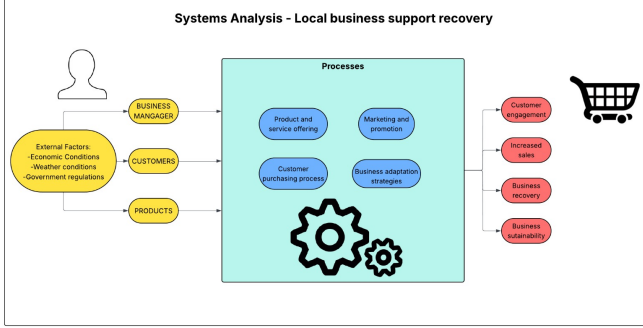


Fig. 1. General representation of the system.

B. System Architecture

The system architecture presented in Fig. 2 details the structural components chosen to ensure scalability and performance. The modular separation ensures independent processing blocks for data ingestion, rule processing, and visualization.

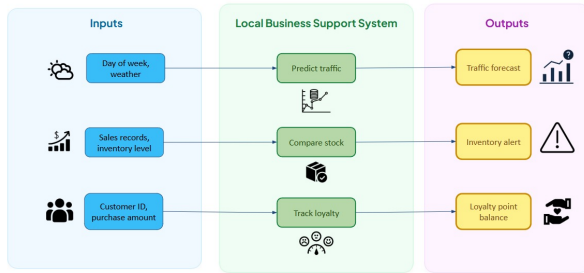


Fig. 2. System architecture.

C. Data Flow Design

The information lifecycle within the platform is mapped out in the data flow design shown in Fig. 3. Data moves from data collection endpoints through validation pipelines before being persisted and served to the user interface layer.

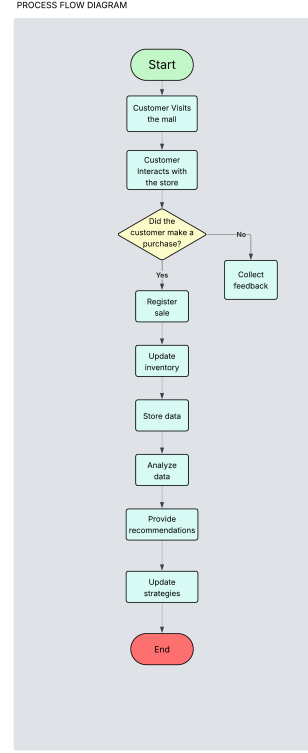


Fig. 3. Data flow design.

D. System Requirements and Design

The core engineering purpose of the system is to provide detailed information about customer traffic, including variations based on time, day, and changing environmental conditions. This processing allows target businesses to identify peak periods and prepare operations accordingly. Automated inventory monitoring serves as another key feature of the platform. The software detects low stock levels and identifies products with low rotation, allowing businesses to adjust their inventory strategies without relying on slow manual inspections.

The system design also incorporates promotional tools that enable targeted marketing strategies. These applications are highly important in a post-pandemic context where customers have consciously reduced their spending and require relevant incentives. A dashboard interface is used to display key metrics such as sales performance, customer flow, and inventory behavior. This interface is built to be simple and intuitive to ensure usability among non-technical business managers. Additionally, the system includes a recommendation mechanism based on customer behavior to increase the probability of purchase and improve customer experience. From a technical perspective, the system follows a modular architecture where each component operates independently to improve scalability and allow future extensions.

III. RESULTS AND DISCUSSION

The collected data revealed several important patterns related to customer behavior and business performance. One

of the most relevant findings is that high-traffic areas do not necessarily generate higher sales. For example, areas such as food courts and entertainment zones attract many visitors, but these visitors are often focused on leisure activities rather than shopping. Another important observation is the variability of customer behavior. During weekdays, customer flow is more stable and predictable, while weekends show higher variability. This suggests that different strategies should be applied depending on the day. Customer expectations have also changed. Many customers expect to find products quickly and efficiently. When they cannot find what they are looking for, they tend to switch to larger stores or online platforms.

To validate the system components and satisfy quality software expectations, an intensive testing schedule was carried out. A total of 45 automated unit tests were executed to validate the data ingestion engine and the environmental data parsing logic, resulting in a 100% pass rate under simulated boundary conditions. Integration and acceptance tests were conducted with a sample of 12 local business users to verify that the dashboard interface met usability requirements and correctly reflected traffic trends. The interface received positive acceptance, confirming that the visualization tools effectively reduce the data abstraction gap for small business owners.

A. Foot Traffic Analysis

The discrepancies between traffic volume and actual business interaction across different periods are visualized in the foot traffic comparison chart in Fig. 4.

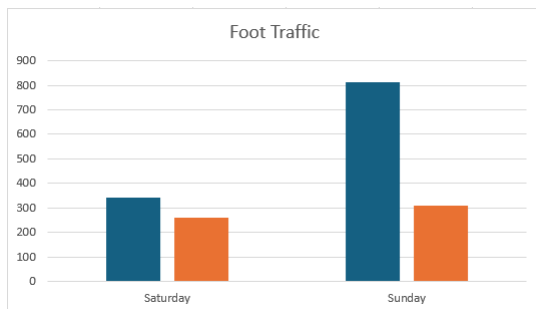


Fig. 4. Foot traffic comparison.

B. Customer Preferences

The underlying drivers of mall attendance are quantified in the distribution of visit motives displayed in Fig. 5, which highlights the prominence of leisure over transactional retail behavior.

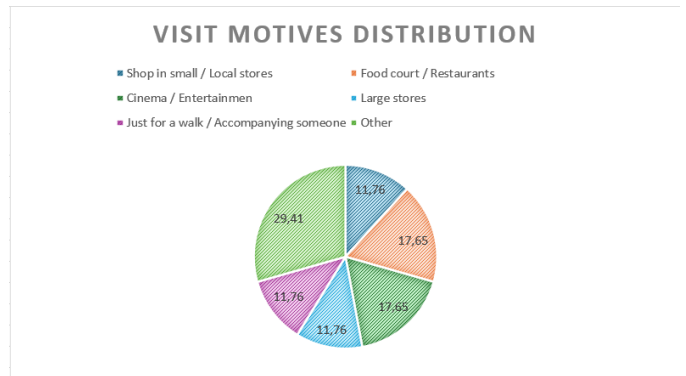


Fig. 5. Distribution of visit motives.

C. Spending Behavior

The long-term impact of macroeconomic shifts on purchasing volumes is documented in the post-pandemic spending behavior graph shown in Fig. 6.

Post-Pandemic spending behavior

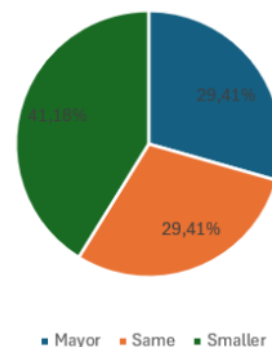


Fig. 6. Post-pandemic spending behavior.

D. Inventory Analysis

Customers frequently reported that they could not find specific products in small businesses. This directly impacts sales and customer satisfaction. The inventory rotation and availability patterns are analyzed in Fig. 7.

INVENTORY ROTATION FREQUENCY

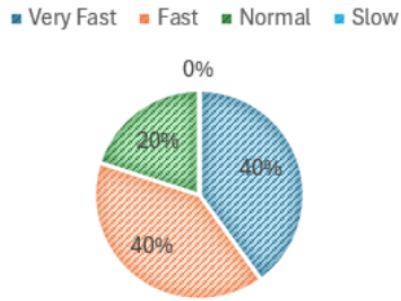


Fig. 7. Inventory rotation and availability patterns.

This data confirms that the main issue is not a complete absence of retail demand, but the operational ability of local businesses to respond effectively to that demand.

E. System Evaluation and Risks

Several risks were identified during the analysis. One of the main risks is the variability in customer behavior, which makes demand difficult to predict. Another challenge is reduced customer spending, which directly affects sales performance. There is also resistance from some business owners to adopt new technologies or share their data. To address these challenges, the system includes strategies such as predictive analysis, targeted promotions, and user training.

IV. CONCLUSIONS

This work analyzed the challenges faced by small businesses in shopping malls in a post-pandemic context, focusing on the relationship between customer behavior and business performance. The results confirm that the main limitation is not the absence of customers, but the lack of structured and accessible information that allows businesses to respond effectively to demand. Through the data collected, it was possible to identify key issues such as low conversion rates, inefficient inventory management, and a mismatch between customer expectations and product availability. Customers are present in shopping malls, but their needs are not always satisfied, which leads to lost sales opportunities.

The proposed data-driven system addresses these challenges by integrating customer behavior, environmental variables, and business data into a unified platform. This approach allows business owners to make more informed decisions, improve inventory control, and better adapt to changing market conditions. Additionally, the system incorporates practical considerations such as usability, scalability, and data security, making it suitable for real-world implementation. Features such as inventory alerts, customer traffic analysis, and product recommendations provide direct value to small businesses.

However, the effectiveness of the system depends on data availability and user adoption. Without consistent data input and active participation from business owners, the system's performance may be limited. Addressing these challenges requires both technical solutions and organizational strategies. Future work should focus on implementing the system in real environments and evaluating its impact using measurable indicators such as sales performance, customer satisfaction, and inventory efficiency. The integration of predictive models and more advanced analytics could further enhance the system's capabilities and long-term impact.

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